Context:
Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.
This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.
By analyzing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil. The information can shed light on various aspects of the business, such as order processing, pricing strategies, payment and shipping efficiency, customer demographics, product characteristics, and customer satisfaction levels.

Dataset:
https://drive.google.com/drive/folders/1TGEc66YKbD443nslRi1bWgVd238gJCnb
The data is available in 8 csv files:

1. **customers.csv**
2. **sellers.csv**
3. **order_items.csv**
4. **geolocation.csv**
5. **payments.csv**
6. **order_reviews.csv**
7. **orders.csv**
8. **products.csv**

The column description for these csv files is given below.

The **customers.csv** contain following features:

Features	Description
customer_id	ID of the consumer who made the purchase
customer_unique_id	Unique ID of the consumer
customer_zip_code_prefix	Zip Code of consumer’s location
customer_city	Name of the City from where order is made
customer_state	State Code from where order is made (Eg. são paulo - SP)

The **sellers.csv** contains following features:

Features	Description
seller_id	Unique ID of the seller registered
seller_zip_code_prefix	Zip Code of the seller's location
seller_city	Name of the City of the seller
seller_state	State Code (Eg. são paulo - SP)

The **order_items.csv** contain following features:

Features	Description
order_id	A Unique ID of order made by the consumers
order_item_id	A Unique ID given to each item ordered in the order
product_id	A Unique ID given to each product available on the site
seller_id	Unique ID of the seller registered in Target
shipping_limit_date	The date before which the ordered product must be shipped
price	Actual price of the products ordered
freight_value	Price rate at which a product is delivered from one point to another

The **geolocations.csv** contain following features:

Features	Description
geolocation_zip_code_prefix	First 5 digits of Zip Code
geolocation_lat	Latitude
geolocation_lng	Longitude
geolocation_city	City
geolocation_state	State

The **payments.csv** contain following features:

Features	Description

order_id	A Unique ID of order made by the consumers
payment_sequential	Sequences of the payments made in case of EMI
payment_type	Mode of payment used (Eg. Credit Card)
payment_installments	Number of installments in case of EMI purchase
payment_value	Total amount paid for the purchase order

The **orders.csv** contain following features:

Features	Description
order_id	A Unique ID of order made by the consumers
customer_id	ID of the consumer who made the purchase
order_status	Status of the order made i.e. delivered, shipped, etc.
order_purchase_timestamp	Timestamp of the purchase
order_delivered_carrier_date	Delivery date at which carrier made the delivery
order_delivered_customer_date	Date at which customer got the product
order_estimated_delivery_date	Estimated delivery date of the products

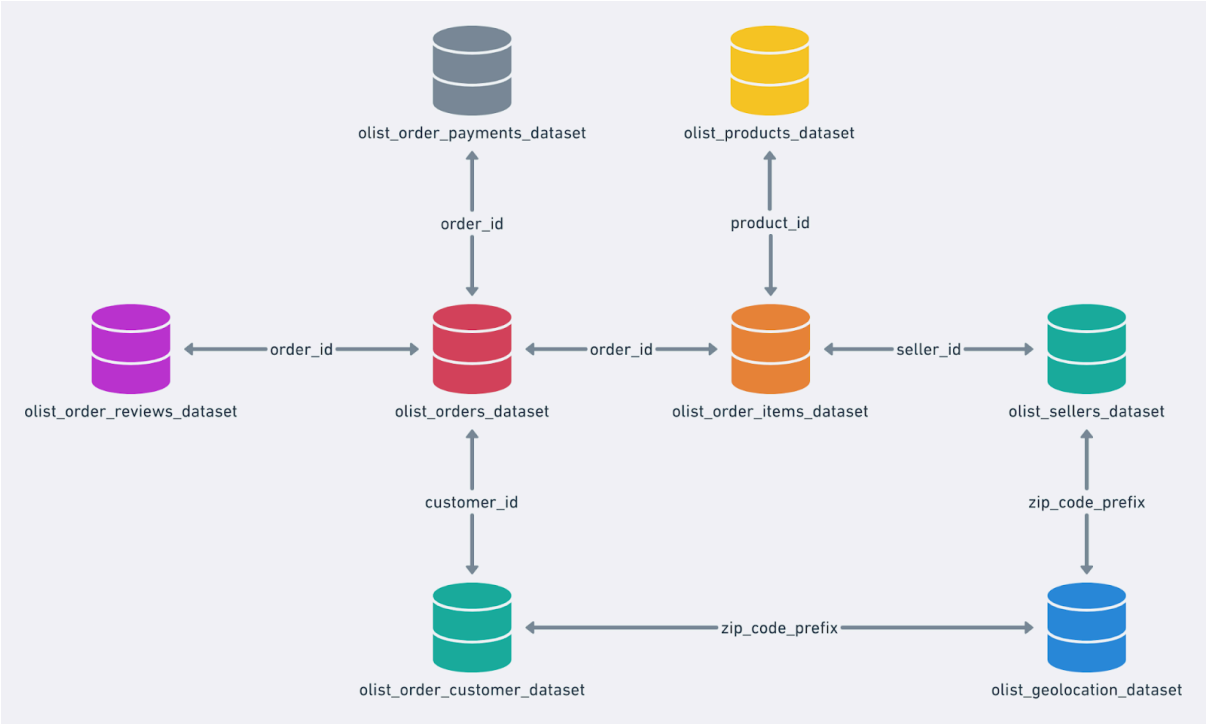
The **order_reviews.csv** contain following features:

Features	Description
review_id	ID of the review given on the product ordered by the order id
order_id	A Unique ID of order made by the consumers
review_score	Review score given by the customer for each order on a scale of 1-5
review_comment_title	Title of the review
review_comment_message	Review comments posted by the consumer for each order
review_creation_date	Timestamp of the review when it is created
review_answer_timestamp	Timestamp of the review answered

The **products.csv** contain following features:

Features	Description
product_id	A Unique identifier for the proposed project.
product_category_name	Name of the product category
product_name_lenght	Length of the string which specifies the name given to the products ordered
product_description_lenght	Length of the description written for each product ordered on the site
product_photos_qty	Number of photos of each product ordered available on the shopping portal
product_weight_g	Weight of the products ordered in grams
product_length_cm	Length of the products ordered in centimeters
product_height_cm	Height of the products ordered in centimeters
product_width_cm	Width of the product ordered in centimeters

Dataset schema:



Problem Statement:
Assuming you are a data analyst/ scientist at Target, you have been assigned the task of analyzing the given dataset to extract valuable insights and provide actionable recommendations.
What does 'good' look like?

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:
    a. Data type of all columns in the "customers" table.
    b. Get the time range between which the orders were placed.
    c. Count the Cities & States of customers who ordered during the given period.
2. In-depth Exploration:
    a. Is there a growing trend in the no. of orders placed over the past years?
    b. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?
    c. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
        i. 0-6 hrs : Dawn
        ii. 7-12 hrs : Mornings
        iii. 13-18 hrs : Afternoon
        iv. 19-23 hrs : Night
3. Evolution of E-commerce orders in the Brazil region:
    a. Get the month on month no. of orders placed in each state.
    b. How are the customers distributed across all the states?
4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.
    a. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
    You can use the "payment_value" column in the payments table to get the cost of orders.
    b. Calculate the Total & Average value of order price for each state.
    c. Calculate the Total & Average value of order freight for each state.
5. Analysis based on sales, freight and delivery time.
    a. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
    Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
    Do this in a single query.

    You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

i. time_to_deliver = order_delivered_customer_date - order_purchase_timestamp
ii. diff_estimated_delivery = order_delivered_customer_date - order_estimated_delivery_date

b. Find out the top 5 states with the highest & lowest average freight value.

c. Find out the top 5 states with the highest & lowest average delivery time.

d. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.
You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

6. Analysis based on the payments:

a. Find the month on month no. of orders placed using different payment types.

b. Find the no. of orders placed on the basis of the payment installments that have been paid.

Evaluation Criteria (100 points):

1. Initial exploration like checking the structure & characteristics of the data (15 points)
2. In-depth Exploration (15 points)
3. Evolution of E-commerce orders in the Brazil region (10 points)
4. Impact on Economy (20 points)
5. Analysis on sales, freight and delivery time (20 points)
6. Analysis based on the payments (10 points)
7. Actionable Insights & Recommendations (10 points)

Submission Process <IMP>:

Once you're done with the case study...

- Use a Word document to paste your SQL queries along with a screenshot of the first 10 rows from the output.
- List down any valuable insights that you find during the analysis and provide some action items from the company's perspective in order to improve the current situation.
- Convert your solutions doc into a PDF, and upload the same on the platform.
- Please note that after submitting once, you will not be allowed to edit your submission.

General Guidelines:

- Evaluation will be kept lenient, so make sure you attempt this case study.
- Try to attempt this before it is discussed in the Live Case Discussion with the Instructor.
- It is understandable that you might struggle with getting started on this or feel stuck at some point.
  In such case:
  - Read the question carefully and try to understand what exactly is being asked.
  - Brainstorm a little. If you're getting an error, remember that Google is your best friend.
  - You can watch the lecture recordings or go through your lecture notes once again if you feel like you're getting confused over some specific topics.
  - Discuss your problems with your peers. Make use of the Slack channel and WhatsApp group.
  - Only if you think that there's a major issue, you can reach out to your Instructor via Slack or Email.

Note: All the solutions are executed in Big Query

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:
   a. Data type of all columns in the "customers" table.
   b. Get the time range between which the orders were placed.
   c. Count the Cities & States of customers who ordered during the given period.

A. Solution 1.a

```
select Column_Name, Data_Type
from `dsml-dec24-case-sql-target.SQL_Target_dataset.INFORMATION_SCHEMA.COLUMNS`
where table_name = 'customers';
```

Query results

JOB INFORMATION | RESULTS | CHART

Row	Column_Name	Data_Type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING

B. Solution 1.b

```
select min(order_purchase_timestamp) as First_Order,
       max(order_purchase_timestamp) as Last_Order
from `SQL_Target_dataset.orders`
```

Query results

JOB INFORMATION | RESULTS | CHART | J

Row	First_Order	Last_Order
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

C. Solution 1.c

```
select count(distinct(customer_city)) as Order_Count_City,
       count(distinct(customer_state)) as Order_Count_State
from `SQL_Target_dataset.customers`
```

Query results

JOB INFORMATION | RESULTS

Row	Order_Count_City	Order_Count_State
1	4119	27

2. In-depth Exploration:
    a. Is there a growing trend in the no. of orders placed over the past years?
    b. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?
    c. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
        i. 0-6 hrs : Dawn
        ii. 7-12 hrs : Mornings
        iii. 13-18 hrs : Afternoon
        iv. 19-23 hrs : Night

A. Solution 2.a

```
with final as
(
select extract (year from order_purchase_timestamp) as Order_Year,
count(order_id) as Total_Orders
from `SQL_Target_dataset.orders`
group by 1
)
select Order_Year, Total_Orders,
    round(
        ((Total_Orders - lag(Total_Orders) over (order by Order_Year)) /
lag(Total_Orders) over (order by Order_Year)) * 100, 2
    ) as Percentage_Change
from final
order by 1
```

Query results

SAVE R

JOB INFORMATION | RESULTS | CHART | JSON

Row	Order_Year	Total_Orders	Percentage_Change
1	2016	329	null
2	2017	45101	13608.51
3	2018	54011	19.76

- **Insights:**
  - The data shows a steady increase in the number of orders over the years, indicating a growing customer base and strong market presence.
  - However, the percentage change in order volume fluctuates, suggesting that external factors like seasonal demand, economic shifts, or promotional campaigns influence purchasing behavior.

- **Recommendations:**
  - Strengthen Logistics & Inventory Management:
    - With growing demand, ensuring that warehousing and fulfillment centers can scale efficiently will be crucial.
  - Investigate Fluctuations:
    - A deeper analysis into the months with significant spikes or dips in orders can help refine promotional strategies.
  - Expand Market Reach:
    - Consider targeted marketing campaigns in regions with lower order volumes to sustain long-term growth.

B. Solution 2.b

```
with final as
(
select extract (year from order_purchase_timestamp) as Order_Year,
      extract (month from order_purchase_timestamp) as Order_Month,
      count(order_id) as Monthly_Orders,
from `SQL_Target_dataset.orders`
group by 2,1
)
select  Order_Month, Order_Year, Monthly_Orders,
    Avg( Monthly_Orders) over (partition by Order_Month order by Order_Month) as
Avg_Monthwise_Orders
from final
order by 1,2
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

ow	Order_Month	Order_Year	Monthly_Orders	Avg_Monthwise_Orders
1	1	2017	800	4034.5
2	1	2018	7269	4034.5
3	2	2017	1780	4254.0
4	2	2018	6728	4254.0
5	3	2017	2682	4946.5
6	3	2018	7211	4946.5
7	4	2017	2404	4671.5
8	4	2018	6939	4671.5
9	5	2017	3700	5286.5
10	5	2018	6873	5286.5

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- **Insights:**
  - Certain months see a higher number of orders, indicating strong seasonal shopping trends.
  - Conversely, some months experience a noticeable dip in order volume, which could be due to a lack of major sales events or changing consumer spending habits.

- **Recommendations:**
  - Align Promotions with High-Demand Months:
    - Running discounts and special offers during peak shopping periods (e.g., Black Friday, holiday sales) can maximize revenue.
  - Boost Off-Season Sales:
    - Implement loyalty programs, limited-time discounts, or product bundling strategies to encourage purchases during slower months.
  - Adjust Inventory Accordingly:
    - Ensuring adequate stock levels during peak months while minimizing excess inventory in slower months will help optimize costs.

C. Solution 2.c

```
select
    case
        when extract(hour from order_purchase_timestamp) between 0 and 6 then
'Dawn'
        when extract(hour from order_purchase_timestamp) between 7 and 12 then
'Morning'
        when extract(hour from order_purchase_timestamp) between 13 and 18 then
'Afternoon'
        else 'Night'
    end as Time_of_Day,
    count(order_id) as Total_Orders
from `SQL_Target_dataset.orders`
group by 1
order by 2 desc;
```

Query results

JOB INFORMATION | RESULTS | CHART

Row	Time_of_Day	Total_Orders
1	Afternoon	38135
2	Night	28331
3	Morning	27733
4	Dawn	5242

- **Insights:**
    - Most orders are placed during the 2nd half of the day, indicating that consumers are more active online after 1pm.
    - Very few purchases are made during the early morning hours, which suggests minimal engagement during this period.

- **Recommendations:**
    - Align Promotions with High-Demand Months:

        - Running discounts and special offers during peak shopping periods (e.g., Black Friday, holiday sales) can maximize revenue.
    - Boost Off-Season Sales:
        - Implement loyalty programs, limited-time discounts, or product bundling strategies to encourage purchases during slower months.
    - Adjust Inventory Accordingly:
        - Ensuring adequate stock levels during peak months while minimizing excess inventory in slower months will help optimize costs.

3. Evolution of E-commerce orders in the Brazil region:
    a. Get the month on month no. of orders placed in each state.
    b. How are the customers distributed across all the states?

A. Solution 3.a

```
with monthly_orders as
(    select
        extract(month from o.order_purchase_timestamp) as Order_Month,
        c.customer_state as Customer_State,
        count(o.order_id) as Total_Orders
    from `dsml-dec24-case-sql-target.SQL_Target_dataset.orders` o
    join `dsml-dec24-case-sql-target.SQL_Target_dataset.customers` c
    on o.customer_id = c.customer_id
    where o.order_status not in ('canceled', 'unavailable')
    group by 1,2
)
select Order_Month, Customer_State,
    sum(Total_Orders) over (partition by Order_Month,Customer_State order by
Customer_State) as Monthly_Orders
from monthly_orders
order by 1, 2;
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

Row	Order_Month	Customer_State	Monthly_Orders
1	1	AC	8
2	1	AL	39
3	1	AM	12
4	1	AP	11
5	1	BA	257
6	1	CE	98
7	1	DF	151
8	1	ES	156
9	1	GO	163
10	1	MA	63

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- **Insights:**
  - Some states show consistent month-over-month growth in orders, while others experience fluctuations.
  - There may be seasonal variations where certain months (e.g., holiday periods, sales events) see a surge in orders.
  - Some states have high order volumes but low month-over-month growth, indicating saturation or untapped expansion potential.

- **Recommendations:**
  - Identify Seasonal Trends:
    - Use historical data to pinpoint peak order months and align marketing campaigns with these trends.
  - Boost Growth in Slow-Growing States:
    - If certain states have low or stagnant growth, consider localized promotions, region-specific discounts, and influencer partnerships.
  - Improve Demand Forecasting:
    - Use AI-driven predictive analytics to anticipate future order trends and adjust inventory levels accordingly.
  - Strengthen Infrastructure in High-Growth States:
    - If certain states are seeing rapid growth, ensure logistics, customer support, and supply chain efficiency keep pace.

B. Solution 3.b

```
select customer_state as Customer_State,
       count(distinct(customer_unique_id)) as No_of_Customers
from `SQL_Target_dataset.customers`
group by 1
order by 2 desc;
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXEC

Row	Customer_State	No_of_Customers
1	SP	40302
2	RJ	12384
3	MG	11259
4	RS	5277
5	PR	4882
6	SC	3534
7	BA	3277
8	DF	2075
9	ES	1964
10	GO	1952

Results per page: 50 1 – 27 of 27

- **Insights:**
  - A few states dominate in terms of customer concentration and order volume, suggesting a stronger e-commerce penetration.
  - Some regions have low customer density, indicating potential market gaps due to infrastructure limitations or lower digital adoption.
  - The disparity in customer distribution suggests that some states may require different engagement strategies.

- **Recommendations:**
  - Expand Market Reach in Underrepresented States:
    - Offer first-time purchase discounts or free shipping to encourage adoption.

        - Improve customer education on e-commerce benefits through localized marketing.
    - Leverage Regional Preferences:
        - Analyze top-selling products in each state and optimize inventory accordingly.
        - Introduce regionalized pricing and promotions based on purchasing power.
    - Enhance Delivery & Service in Low-Customer Regions:
        - If slow adoption is due to high shipping costs or delivery delays, explore new warehouse locations or local courier partnerships.
        - Offer cash-on-delivery (COD) in regions with lower credit card penetration to increase trust in online shopping.

4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.
    a. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
       You can use the "payment_value" column in the payments table to get the cost of orders.
    b. Calculate the Total & Average value of order price for each state.
    c. Calculate the Total & Average value of order freight for each state.

A. Solution 4.a

```
with order_cost as
(select extract(year from o.order_purchase_timestamp) as Order_Year,
        sum(p.payment_value) as Total_Payment
from `dsml-dec24-case-sql-target.SQL_Target_dataset.orders` o
join `dsml-dec24-case-sql-target.SQL_Target_dataset.payments` p
on o.order_id = p.order_id
where   extract(month from o.order_purchase_timestamp) between 1 and 8
        and extract(year from o.order_purchase_timestamp)  between 2017 and 2018
        and o.order_status not in ('canceled', 'unavailable')  -- Excluding
canceled/unavailable orders
    group by Order_Year
)
select Order_Year,Total_Payment,
    lag(Total_Payment) over (order by Order_Year) as Prev_Year_Payment,
    case
        when lag(Total_Payment) over (order by Order_Year) is not null
```

```
        then round(((Total_Payment - lag(Total_Payment) over (order by
Order_Year)) / lag(Total_Payment) over (order by Order_Year)) * 100, 2)
        else null
    end as Percent_Increase
from order_cost;
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECU

Row	Order_Year	Total_Payment	Prev_Year_Payment	Percent_Increase
1	2018	8594665.519999...	3575957.459999...	140.35
2	2017	3575957.459999...	null	null

Results per page: 50 1 – 2 of 2

B. Solution 4.b

```
with final as (
select c.customer_state as State,
sum(price) as Total_Price,
count(distinct(o.order_id)) as No_of_Orders
from `SQL_Target_dataset.orders` o
inner join `SQL_Target_dataset.order_items` i
on o.order_id= i.order_id
inner join `SQL_Target_dataset.customers` c
on o.customer_id=c.customer_id
where o.order_status not in ('canceled', 'unavailable')
group by state
)
select State,
Total_Price,
No_of_Orders,
(Total_Price/No_of_Orders) as Avg_Price
from final
order by Total_Price desc
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON

Row	State	Total_Price	No_of_Orders	Avg_Price
1	SP	5163867.220001...	41125	125.5651603...
2	RJ	1811623.419999...	12697	142.6812176...
3	MG	1573508.199999...	11496	136.8744084...
4	RS	742559.7800000...	5415	137.1301532...
5	PR	676883.0600000...	4982	135.8657286...
6	SC	518578.2800000...	3599	144.0895470...
7	BA	507108.8300000...	3344	151.6473773...
8	DF	300886.4499999...	2120	141.9275707...
9	GO	287870.4599999...	1998	144.0793093...
10	ES	273532.1299999...	2018	135.5461496...

Results per page: 50 1 – 27 of 27

C. Solution 4.c

```
with final as (
select c.customer_state as State,
sum(freight_value) as Total_Freight_Value,
count(distinct(o.order_id)) as No_of_Orders
from `SQL_Target_dataset.orders` o
inner join `SQL_Target_dataset.order_items` i
on o.order_id= i.order_id
inner join `SQL_Target_dataset.customers` c
on o.customer_id=c.customer_id
where o.order_status not in ('canceled', 'unavailable')
group by State
)
select State,
Total_Freight_Value,
No_of_Orders,
(Total_Freight_Value/No_of_Orders) as Avg_Freight_Value
from final
order by Total_Freight_Value desc
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON

Row	State	Total_Freight_Value	No_of_Orders	Avg_Freight_Value
1	SP	714264.8399999...	41125	17.368142006...
2	RJ	304044.1400000...	12697	23.946140033...
3	MG	269566.2300000...	11496	23.448697807...
4	RS	134730.8100000...	5415	24.881036011...
5	PR	117313.5500000...	4982	23.547480931...
6	BA	99799.82999999...	3344	29.844446770...
7	SC	89445.42000000...	3599	24.852853570...
8	PE	59278.39999999...	1643	36.079367011...
9	GO	52673.90999999...	1998	26.363318318...
10	DF	50440.75999999...	2120	23.792811320...

Results per page: 50 1 – 27 of 27

- **Insights:**
  - The total payment value for orders increased from 2017 to 2018, indicating higher consumer spending.
  - However, the rate of increase suggests a potential rise in product prices, shipping fees, or higher-value purchases.
  - The average cost of orders does not increase uniformly across all states, meaning some regions contribute more to revenue growth than others.
  - Certain states contribute significantly higher total order value, while others have low average order values, indicating disparities in purchasing power.
  - High-value states are economically stronger or have better access to e-commerce services.
  - Some states may have high total order value but lower average order value, suggesting a large customer base making frequent, lower-cost purchases.
  - Freight costs vary significantly across states, with some states having disproportionately high shipping costs.

    - States farther from warehouses or with weak transportation infrastructure experience higher freight costs, impacting overall order costs.
    - High shipping costs might deter customers from placing larger orders, especially in states with low total order values.

- **Recommendations:**
    - Optimize Pricing Strategies:
        - Analyze whether the increase in order costs is driven by inflation, higher-value products, or strategic pricing changes.
        - Adjust pricing models accordingly.
    - Expand High-Performing Regions:
        - Identify the states contributing most to the revenue increase and focus on expanding marketing efforts and logistics in these regions.
    - Introduce Flexible Payment Options:
        - Offering EMI, Buy Now Pay Later (BNPL), and discount-based prepaid orders can help retain price-sensitive customers.
    - Customize Marketing Strategies by Region:
        - In high-value states, promote premium products and exclusive deals.
        - In low-value states, offer discounts, bulk purchases, and affordable product bundles.
    - Improve Product Offerings:
        - Analyze the product preferences in high- and low-spending states and tailor inventory to suit demand.
    - Enhance Logistics in Low-Spending Areas:
        - If lower spending is due to shipping costs, introduce subsidized shipping, free delivery thresholds, or local fulfillment centers.
    - Optimize Warehouse Locations:
        - Open regional distribution hubs to reduce long-distance shipping costs.
    - Partner with Local Delivery Services:
        - In high-freight-cost states, third-party courier partnerships can reduce shipping expenses.
    - Introduce Free Shipping Thresholds:
        - Encourage customers to increase order value to qualify for free or discounted shipping.

5. Analysis based on sales, freight and delivery time.

a. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
   Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
   Do this in a single query. You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:
   i. time_to_deliver = order_delivered_customer_date - order_purchase_timestamp
   ii. diff_estimated_delivery = order_delivered_customer_date - order_estimated_delivery_date
b. Find out the top 5 states with the highest & lowest average freight value.
c. Find out the top 5 states with the highest & lowest average delivery time.
d. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.
   You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

A. Solution 5.a

```
select order_id,
    date_diff(order_delivered_customer_date, order_purchase_timestamp, DAY) as
time_to_deliver,
    date_diff(order_delivered_customer_date, order_estimated_delivery_date, DAY)
as diff_estimated_delivery
from `SQL_Target_dataset.orders`
where order_status = 'delivered';
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON

Row	order_id	time_to_deliver	diff_estimated_delivery
1	635c894d068ac37e6e03...	30	-1
2	3b97562c3aee8bdedcb5...	32	0
3	68f47f50f04c4cb677457...	29	-1
4	276e9ec344d3bf029ff83...	43	4
5	54e1a3c2b97fb0809da5...	40	4
6	fd04fa4105ee8045f6a01...	37	1
7	302bb8109d097a9fc6e9...	33	5
8	66057d37308e787052a3...	38	6
9	19135c945c554eebfd75...	36	2
10	4493e45e7ca1084efcd3...	34	0

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- **Insights:**
  - The average time to deliver orders varies significantly across customers, with some experiencing much longer wait times.
  - The difference between estimated and actual delivery dates shows that some customers consistently experience delays (shown with +ve values in the last column), while others receive deliveries earlier than expected (shown with -ve values in the last column).
  - Longer delivery times are likely due to geographical distance, logistical inefficiencies, or warehouse distribution gaps.

- **Recommendations:**
  - Improve Last-Mile Delivery:
    - Partner with regional logistics providers or use local fulfillment centers to minimize delivery times.
  - Enhance Estimated Delivery Accuracy:
    - Improve machine-learning-based demand forecasting to ensure realistic estimated delivery dates.
  - Implement Express Shipping for High-Demand Regions:

- Offer paid express delivery in states where customers frequently experience delays.

B. Solution 5.b

```sql
with final as (
    select
        c.customer_state as State,
        sum(freight_value) as Total_Freight_Value,
        count(distinct(o.order_id)) as No_of_Orders
    from `SQL_Target_dataset.orders` o
    inner join `SQL_Target_dataset.customers` c on o.customer_id=c.customer_id
    inner join `SQL_Target_dataset.order_items` i on o.order_id= i.order_id
    group by 1),
top_five as (
select State,
       (Total_Freight_Value/No_of_Orders) as Avg_Freight_Value
from final
order by Avg_Freight_Value desc
Limit 5),
lowest_five as (
select State,
       (Total_Freight_Value/No_of_Orders) as Avg_Freight_Value
from final
order by Avg_Freight_Value asc
Limit 5)

select* from lowest_five
UNION ALL
select* from top_five;
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

Row	State	Avg_Freight_Value
1	RR	48.59108695652...
2	PB	48.34535714285...
3	RO	46.22421052631...
4	AC	45.51543209876...
5	PI	43.03894523326...
6	SP	17.37095033232...
7	MG	23.46270443520...
8	PR	23.57976790716...
9	DF	23.82376470588...
10	RJ	23.94525231154...

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- **Insights:**
  - Some states have significantly higher average freight costs, possibly due to remote locations or inefficient shipping routes (1st five rows shows the highest 5 average freight rates).
  - States with the lowest freight values likely have better infrastructure, warehouse proximity, or high order density, reducing per-order shipping costs (last five rows shows the lowest 5 average freight rates).

- **Recommendations:**
  - Optimize Warehouse Locations:
    - Establish new distribution hubs closer to high-freight-cost states to minimize shipping expenses.
  - Negotiate Carrier Rates:
    - Work with logistics partners to secure discounted bulk shipping rates in high-cost areas.
  - Introduce Free Shipping Thresholds:
    - Encourage customers to increase cart value to qualify for free or reduced shipping fees.

C. Solution 5.c

```sql
with final as (
    select c.customer_state as State,
avg(DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY))
as Avg_Delivery_Time,
        count(distinct(o.order_id)) as No_of_Orders
    from `SQL_Target_dataset.orders` o
    inner join `SQL_Target_dataset.customers` c on o.customer_id = c.customer_id
    where o.order_status = 'delivered'  -- Exclude invalid orders
    group by c.customer_state
),

top_five as
(select State,
         Avg_Delivery_Time
from final
order by Avg_Delivery_Time desc
Limit 5),

lowest_five as
(select State,
        Avg_Delivery_Time
from final
order by Avg_Delivery_Time asc
Limit 5)

select* from lowest_five
UNION ALL
select* from top_five;
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON

Row	State	Avg_Delivery_Time
1	RR	28.97560975609...
2	AP	26.73134328358...
3	AM	25.98620689655...
4	AL	24.04030226700...
5	PA	23.31606765327...
6	SP	8.298093544722...
7	PR	11.52671135486...
8	MG	11.54218777523...
9	DF	12.50913461538...
10	SC	14.47518330513...

- **Insights:**
  - States with the longest delivery times are likely geographically distant or underserved by logistics partners (1st five rows show the highest 5 average delivery times).
  - The fastest delivery states may have better connectivity due to urban infrastructure or local warehouse presence (last five rows show the lowest 5 average delivery times).

- **Recommendations:**
  - Invest in Regional Distribution:
    - Expand warehousing and micro-fulfillment centers in high-delay regions to reduce shipping distances.
  - Improve Route Planning:
    - Use AI-powered logistics tools to dynamically optimize delivery routes and prioritize faster shipments.
  - Offer Same-Day/Next-Day Delivery in Key Locations:

- If feasible, launch priority shipping services in states where infrastructure allows for faster deliveries.

D. Solution 5.d

```
WITH state_delivery_time AS (
    SELECT
        c.customer_state AS State,
        AVG(DATE_DIFF(o.order_estimated_delivery_date,
o.order_delivered_customer_date, DAY)) AS Avg_Delivery_Early
    FROM `SQL_Target_dataset.orders` o
    INNER JOIN `SQL_Target_dataset.customers` c
        ON o.customer_id = c.customer_id
    WHERE o.order_status = 'delivered'  -- Consider only delivered orders
    GROUP BY c.customer_state  -- Grouping by state
)
SELECT State, Avg_Delivery_Early
FROM state_delivery_time
ORDER BY Avg_Delivery_Early DESC
LIMIT 5;
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON

Row	State	Avg_Delivery_Early
1	AC	19.76250000000...
2	RO	19.13168724279...
3	AP	18.73134328358...
4	AM	18.60689655172...
5	RR	16.41463414634...

- **Insights:**
  - Some states consistently receive orders earlier than the estimated delivery date, indicating efficient logistics and distribution networks.

    - Customers in these states may develop higher service expectations, expecting rapid shipping for all purchases.

- **Recommendations:**
    - Use Early Delivery as a Competitive Advantage:
        - Promote "Fast Delivery Promise" marketing campaigns in these regions.
    - Replicate Best Practices in Other States:
        - Study why these states perform well (better warehouse placement, faster shipping methods) and apply similar strategies elsewhere.
    - Introduce Subscription-Based Priority Shipping:
        - Consider a premium membership program (e.g., Amazon Prime-like model) for guaranteed fast shipping.

6. Analysis based on the payments:
    a. Find the month on month no. of orders placed using different payment types.
    b. Find the no. of orders placed on the basis of the payment installments that have been paid.

A. Solution 6.a

```
with monthly_orders as
(
    select extract(month from o.order_purchase_timestamp) as Order_Month,
           p.payment_type as Payment_Type,
           count(o.order_id) as Total_Orders
    from `dsml-dec24-case-sql-target.SQL_Target_dataset.orders` o
    join `dsml-dec24-case-sql-target.SQL_Target_dataset.payments` p
    on o.order_id = p.order_id
    where o.order_status not in ('canceled', 'unavailable')
    group by Order_Month, Payment_Type
)
select Order_Month, Payment_Type,
    sum(Total_Orders) over (partition by Order_Month,Payment_Type order by
Payment_Type) as Monthly_Orders
from monthly_orders
order by 1, 2;
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXE

Row	Order_Month	Payment_Type	Monthly_Orders
1	1	UPI	1692
2	1	credit_card	6031
3	1	debit_card	118
4	1	voucher	472
5	2	UPI	1694
6	2	credit_card	6476
7	2	debit_card	81
8	2	voucher	416
9	3	UPI	1925
10	3	credit_card	7617

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- **Insights:**
  - Credit cards dominate transactions, indicating customer preference for convenience and flexibility.
  - UPI, vouchers and debit card payments have a stable but lower share, suggesting that some customers still prefer non-credit options.
  - Seasonal variations exist in payment preferences, with installment-based payments increasing during peak shopping periods (e.g., Black Friday, holiday sales).

- **Recommendations:**
  - Encourage Digital Wallet & Instant Payment Adoption:
    - Promote the use of faster payment options (e.g., UPI, Google Pay, Apple Pay) through discounts or cashback offers.
  - Optimize Checkout Experience for Preferred Payment Methods:
    - Ensure that credit cards, UPI and digital wallets are prominently displayed at checkout.
  - Leverage Payment Data for Targeted Marketing:
    - Offer personalized promotions based on a customer's preferred payment method.
  - Reduce Cart Abandonment with Payment Reminders:
    - Use email/SMS reminders for pending payments to improve conversion rates.

B. Solution 6.b

```sql
select
  payment_installments as Installments_Paid,
count(order_id) as No_Orders,
from `dsml-dec24-case-sql-target.SQL_Target_dataset.payments`
where payment_installments >= 1
group by payment_installments
order by No_Orders desc
```

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART

Row	Installments_Paid	No_Orders
1	1	52546
2	2	12413
3	3	10461
4	4	7098
5	10	5328
6	5	5239
7	8	4268
8	6	3920
9	7	1626
10	9	644

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- **Insights:**
  - A significant number of customers prefer installment-based purchases, particularly for higher-value products.
  - Customers opting for higher installment plans (6+ months) may be more price-sensitive, meaning they are likely to respond well to promotional financing.
  - Shorter installment plans (1-3 months) indicate customers willing to pay faster but still prefer flexible payments.

- **Recommendations:**
  - Offer Interest-Free Installments for Higher Orders:
    - Encourage larger purchases by providing zero-interest EMI options on high-ticket items.
  - Segment & Target High-Installment Customers:
    - Customers opting for longer installment plans (6+ months) can be targeted with personalized discounts or loyalty rewards.
  - Enhance Transparency in Installment Plans:
    - Clearly display total interest rates and installment breakdowns at checkout to reduce confusion.
  - Introduce Pay-Later Options:
    - Partner with BNPL (Buy Now, Pay Later) providers to attract more budget-conscious customers.